MLCC 2026

Lab sessions

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About the labs

- Monday, Tuesday and Thursday 2.30-4.30pm
- Python notebooks – Colab is recommended
- https://github.com/mletizia/mlcc-2026

About the labs

```python
def euclidDistance(P1,P2):
    return np.linalg.norm(P1-P2,2)
```

... and then a function that computes all the distances between two set of points stored in two matrices.

```python
def allDistances(X1, X2):
    D = np.zeros((X1.shape[0], X2.shape[0]))
    # TODO: Implement the computation of the pairwise Euclidean distances
    # Expected output:
    # D is a (n1 × n2) array, where D[i, j] equals the Euclidean distance
    # between the i-th sample in X1 and the j-th sample in X2.
    return D
```

Cross validation and Noise in k-NN

In this final task we consider the effect of noise on the best k (chosen with k-fold CV).

We will use dataset3 which has high noise (20%), and dataset4 which has low noise (5%).

You will have to do the following:

1. Use k-fold CV to find the best k for datasets 3 and 4.
2. Compare the obtained values of k, with the actual best k on the test error of the two datasets. This should be a sanity check: if the errors on k-fold CV and on the test set are very different, something may be wrong with your code.
3. Comment on how noise affects the best k: does a more noisy dataset need a higher or a lower k, and why? It may be useful to plot the separating function (using the `separatingFkNN` function) of k-NN with the best k for the two datasets.

```python
# ... your code goes here... #
```

Python

Today

K-Nearest Neighbours and Cross Validation

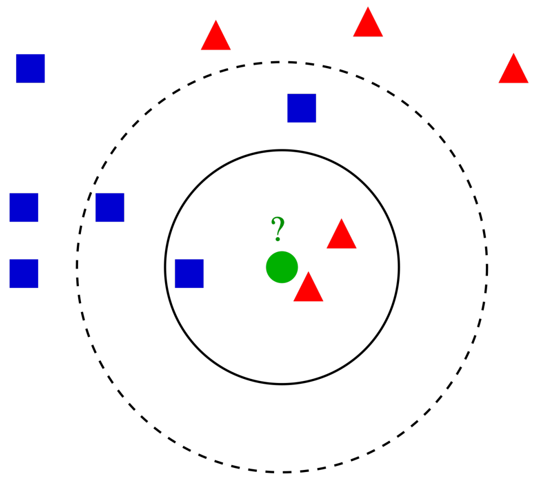
- KNN is a nonparametric method for supervised learning

Local method (not ERM):

output is based on local information → nearby inputs have the same output

Mostly used for classification

$$\hat{f}(x_{\text{new}}) = \text{majority}\{y_i | i \in \widehat{N}_K(x_{\text{new}})\}$$



From Wikipedia

Today

K-Nearest Neighbours and Cross Validation

- KNN is a nonparametric method for supervised learning
  - Generate/load data
  - Define the KNN classifier
  - Train and evaluate the classifier at varying K, with and without noise
  - Visualize decision boundary and plot the loss

Today

K-Nearest Neighbours and Cross Validation

- V-fold Cross Validation: *how to choose K in practice?*